A multimodal learning analytics pipeline for collaborative making-based learning

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Abstract. Identifying collaborative making-based conceptual learning processes remains an analytical challenge. We contribute a fully computational multimodal learning analytics pipeline that 1) leverages computer vision and machine learning to extract, from participants audio and video, theoretically grounded behavioural features, 2) relates these features to students learning gains to construct behavioural profiles of collaborative making-based conceptual learning.

Keywords: Collaborative making-based learning · multimodal learning analytics.

1 Introduction and Background

Making-based learning is an embodied, multimodal and collaborative process [7]. The nature of actions while making can discriminate learners from non-learners [7] and expertise [1]. In fact, even the quantity and speed of movement of certain hand landmarks differentiates learners from non-learners [4]. When students collaborate, dialogue quality such as taking turns and transactivity, and speech features such as pitch and loudness, can both be indicative of good collaboration [2]. Joint visual attention is used to build shared understanding in collaboration, and so associated with better collaboration quality and learning gains [5]. Finally, arousal and valence are used to indicate emotional state in collaborative learning [2]. Together, literature suggests that analysing collaborative making-based conceptual learning processes comprehensively requires considering individual and collaborative actions or movement, speech, eye gaze and affect. Such an analysis is challenging, and so we ask this research question: *How can multimodal behavioural features be computationally extracted from audio and video recordings to understand collaborative making-based conceptual learning processes?* While MMLA has been used to evaluate collaboration in hands-on learning situations, MMLA of collaborative making-based conceptual learning is unexplored. In this work, we contribute a fully computational MMLA pipeline to assess collaborative making-based conceptual learning.

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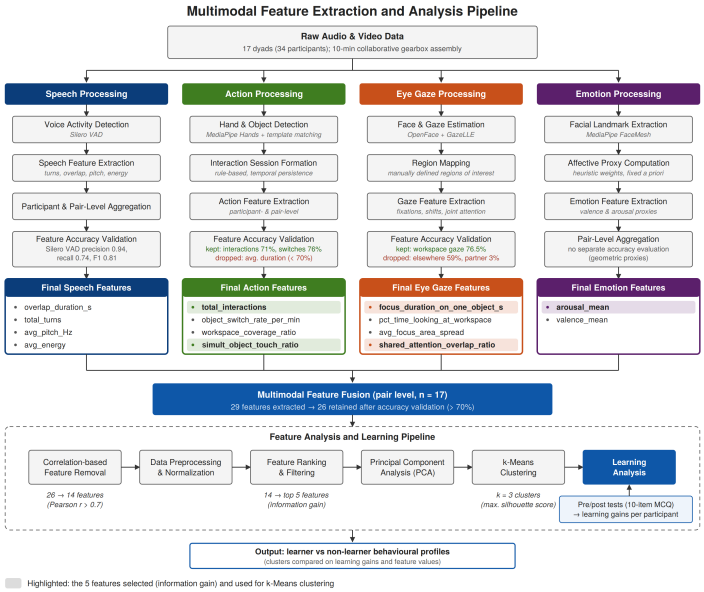
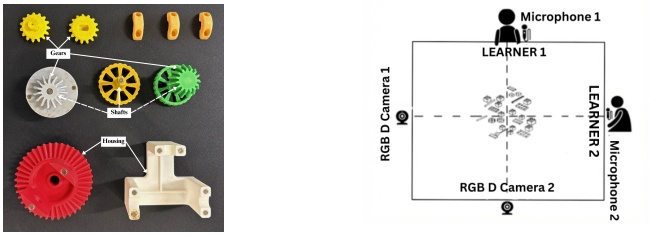


Fig. 1. MMLA pipeline using behaviours

2 Methodology

MMLA pipeline: We propose a MMLA pipeline (Figure 1) to relate our sets of theory-inspired multimodal features to learning outcomes: actions, speech, eye gaze and affect. **Actions:** Hands are tracked with MediaPipe Hands and assigned to participants by a fixed image partition. Task objects are detected by colour and size templates. A touch counted as an interaction only when hand–object contact persisted over consecutive frames, and touches were merged into sessions with start and end times. Six pair-level features were derived, covering interaction frequency, object switching, spatial coverage and simultaneous object handling. **Speech:** Speech activity is detected separately for each participant using Silero VAD [6], which returns start–end intervals from which all features were computed. **Eye gaze:** Gaze is estimated using GazeLLE [3], with OpenFace for face localisation. Gaze points taken from GazeLLE's probabilistic heatmaps are mapped to task regions (workspace, partner, elsewhere) using the fixed camera and task layout. Features describing fixations, gaze shifts, focus spread and shared attention are computed from the smoothed trajectories and heatmaps. **Affect:** Per-frame MediaPipe FaceMesh landmarks are normalised by inter-ocular distance. Geometric features associated with affect (eye opening, mouth opening, mouth width, brow separation, lip compression) are smoothed and scaled to 0–1 within each recording, then linearly combined with heuristic weights into arousal and valence proxies of relative facial activity.



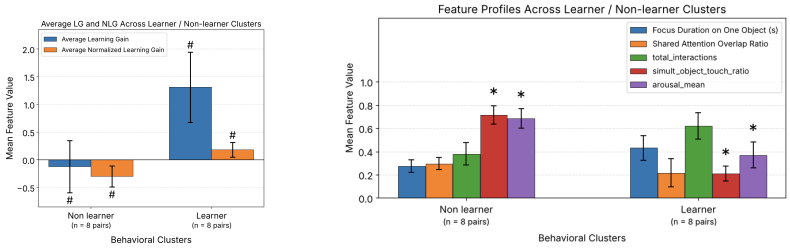
(a) Gearbox parts (b) Experimental set-up

Fig. 2. The making task and experimental set-up

Data Collection: We tested the pipeline in a pilot study; 17 dyads of second or third year undergraduate engineering students, recruited according to Institutional Ethics Committee guidelines (with written informed consent), participated. The study task, taken from [1], required participants to assemble a differential gearbox from ten magnet-embedded pieces (Figure 2a) without any instructions, diagrams, or hints. The protocol consisted of three 10-minute stages: pre-test, making-task, and post-test. A 10-item MCQ test, similar to one used in [4]), was used to assess participants' conceptual understanding about differential gears. Participants did the tests individually, while collaboration was encouraged during the making task. The task was recorded using three high-quality cameras: a Brio webcam positioned above the workspace to capture participants' hands, and two Intel RealSense D435i RGB-D cameras positioned in front of each participant to capture their faces and part of the workspace. Sensitive Lavalier microphones attached to each participant captured speech data (Figure 2b).

3 Findings and Conclusion

Action and eye gaze features defined contextually were validated against manual annotation of randomly sampled 30-second segments (5% of the data) to assess accuracy. Speech activity was compared with WebRTC VAD (modes 2 and 3) and an energy-based detector for agreement (mean precision 0.94, recall 0.74, F1 0.81). Affect features' accuracy was not separately evaluated as they are geometric rather than categorical. Overall, 26 of 29 features with accuracy $> 70\%$ were retained (Figure 1) and 14 after correlation analysis (Pearson's $r > 0.7$). Next, information gain for splitting learners versus non-learners (mean learning gain > 0 or ≤ 0 respectively) was computed for each feature, and the top five were selected for k-means clustering; $k = 3$ clusters were chosen for highest silhouette score (0.28). One cluster with a single pair was excluded, and the learning gains and feature values of the remaining two clusters ($N = 8$ each) were statistically compared (Fig. 3). Learners have significantly lower *simult_object_touch_ratio* and *arousal_mean*, suggesting fewer instances of simultaneously handling the same object and a calmer affect, similar to previous work [2, 7]. Learners also



(a) Learning gains (b) Behavioural profiles

Fig. 3. Learning and behavioural profiles. $*$: $p < 0.05$, #: $0.05 < p < 0.1$

interacted more with the objects, spent a longer time focused on one object and had lower shared attention duration (not statistically different).This last finding contradicts previous research [5] and merits further investigation. Together, a learning profile is marked by increased interactions, a longer individual gaze focus, looking at different parts of the workspace and handling different objects, and a calmer emotional state. In conclusion, our video and audio-based fully computational MMLA pipeline to characterize collaborative making-based conceptual learning processes, and these preliminary findings, lay the foundation for future research in collaborative making-based learning.

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